

EXAMPLE 1

Enter the starting number:

[illegible]

Step 1 (current number):

[illegible]

-> After squaring digits:

[illegible]

Step 2 (current number):

[illegible]

-> After squaring digits:

[illegible]

[illegible]

Step 3 (current number):

[illegible]

-> After squaring digits:

[illegible]

4113664113664113664113664113664113664113664113664113664113664113664113664113
6641136641136641136641136641136641136641136641136641136641136641136641136641
136641136641136641136641136641136641136641136641136641

-> After summing groups of 3:

[illegible]

Step 4 (current number):

[illegible]

-> After squaring digits:

[illegible]

Brutal stable number reached:

EXAMPLE 2

Enter the starting number:

Step 1 (current number):

[illegible]

[illegible]

813813813813813813813813813813813813131416141614161416141614161416
141614161416141614161416141614162227272727272727272212121212121212

813813813813813813813813813813813813131416141614161416141614161416
141614161416141614161416141614162227272727272727272212121212121212

[illegible]

64196419641964196419641964196419641964196419641919116136116136116136
1161361161361161361161361161361161361161361161361161361161361161361161
3644449449449449449449449449449449441414141414141414

64196419641964196419641964196419641964196419641919116136116136116136
1161361161361161361161361161361161361161361161361161361161361161361161
3644449449449449449449449449449449441414141414141414

[illegible]

1015108121310101510812131010151081213101015108121310101510810101013810185161
0101381018516101013810185161010138101851610101381018516101013138131013101310
1310131013101310131013101310138888888888

2611517962694426115179121291224914811221472129122491481512454245424542412242
424

Step 4 (current number):

2611517962694426115179121291224914811221472129122491481512454245424542412242
424

-> After squaring digits:

4361125149813643681161643611251498114148114416811166411441164941481144168111
664125141625164162516416251641614416416416

Step 5 (current number):

4361125149813643681161643611251498114148114416811166411441164941481144168111
664125141625164162516416251641614416416416

-> After squaring digits:

1693611425116816419361693664113613616936114251168164111611664111616136641113
6361611161611361681161166411161613664111363616142511613642513616136425136161
3642513616136116161361613616136

-> After summing groups of 3:

1610771511181615610101881181568136810163121331351310816313101151581181011108
138101012913108810101310

Step 6 (current number):

1610771511181615610101881181568136810163121331351310816313101151581181011108
138101012913108810101310

-> After squaring digits:

1361049491251116413612536101016464116412536641936641013691419919251910641369
19101125125641164101110641964101014811910646410101910

-> After summing groups of 3:

105228311101072166111016131621861916111010192881111221119251010165210

-> After summing groups of 3: 6125291337591516311217351181123

Step 7 (current number): 6125291337591516311217351181123

-> After squaring digits: 361425481199492581125136911414992511641149

-> After summing groups of 3: 1011131915148101192071114

Step 8 (current number): 1011131915148101192071114

-> After squaring digits: 10111918112511664101181404911116

Step 9 (current number): 10111918112511664101181404911116

-> After squaring digits: 1011181164114251136361610116411601681111136

Step 10 (current number): 1011181164114251136361610116411601681111136

-> After squaring digits: 101116411361611164251193693613610113616113601366411111936

-> After summing groups of 3: 286108118131810751357156318

-> After summing groups of 3: 169105913151212

-> After summing groups of 3: 1661375

Step 11 (current number): 1661375

-> After squaring digits: 13636194925

Step 12 (current number): 13636194925

-> After squaring digits: 19369361811681425

Step 13 (current number): 19369361811681425

-> After squaring digits: 18193681936164113664116425

Step 14 (current number): 18193681936164113664116425

-> After squaring digits: 1641819366418193613616119363616113616425

Step 15 (current number): 1641819366418193613616119363616113616425

-> After squaring digits:

136161641819363616164181936193613611819369361361193613616425

-> After summing groups of 3: 10811181213111018131081818101013101311

Step 16 (current number): 10811181213111018131081818101013101311

-> After squaring digits: 10641116414191110164191064164164101019101911

Step 17 (current number): 10641116414191110164191064164164101019101911

-> After squaring digits:

1036161113616116181111013616181103616136161361610101811018111

Step 18 (current number):

1036161113616116181111013616181103616136161361610101811018111

-> After squaring digits:

109361361119361361136164111101936136164110936136193613619361361010164110164111

-> After summing groups of 3: 1010101110101011321810112181013101610101112113

Step 19 (current number): 1010101110101011321810112181013101610101112113

-> After squaring digits: 1010101110101011941641011416410191013610101114119

Step 20 (current number): 1010101110101011941641011416410191013610101114119

-> After squaring digits:

101010111010101181161361610111613616101811019361010111161181

-> After summing groups of 3: 2131210810731013210101013810

Step 21 (current number): 2131210810731013210101013810

-> After squaring digits: 4191410641049910194101010196410

Step 22 (current number): 4191410641049910194101010196410

-> After squaring digits: 16181116103616101681811018116101010181361610

Step 23 (current number): 16181116103616101681811018116101010181361610

-> After squaring digits: 136164111361093613610136641641101641136101010164193613610

-> After summing groups of 3: 10113101210710111121110211113107

Step 24 (current number): 10113101210710111121110211113107

-> After squaring digits: 1011910141049101111411104111191049

Step 25 (current number): 1011910141049101111411104111191049

-> After squaring digits: 1011811011610168110111116111016111181101681

Step 26 (current number): 1011811011610168110111116111016111181101681

-> After squaring digits: 1011641101136101366411011111361110136111164110136641

Step 27 (current number): 1011641101136101366411011111361110136111164110136641

-> After squaring digits:

10113616110119361019363616110111119361110193611113616110193636161

Step 28 (current number):

10113616110119361019363616110111119361110193611113616110193636161

-> After squaring digits:

1011936136110118193610181936936136110111118193611101819361111936136110181936
9361361

Step 29 (current number):

1011936136110118193610181936936136110111118193611101819361111936136110181936
9361361

-> After squaring digits:

1011819361936110116418193610164181936819361936110111116418193611101641819361
1118193619361101641819368193619361

Step 30 (current number):

1011819361936110116418193610164181936819361936110111116418193611101641819361
1118193619361101641819368193619361

-> After squaring digits:

1011641819361819361101136161641819361013616164181936641819361819361101111136
1616418193611101361616418193611116418193618193611013616164181936641819361819
361

Step 31 (current number):

1011641819361819361101136161641819361013616164181936641819361819361101111136
1616418193611101361616418193611116418193618193611013616164181936641819361819
361

-> After squaring digits:

1011361616418193616418193611011936136136161641819361019361361361616418193636
1616418193616418193611011111936136136161641819361110193613613616164181936111
1361616418193616418193611019361361361616418193636161641819361641819361

-> After summing groups of 3:

2108111810111810213101013111018218101081118121311101811101823111010101313138
218101081118103101313131313810101010131313158111810111810

-> After summing groups of 3:

310102103425211993115210211521571199394575792157143939

-> After summing groups of 3: 434114217348131916171813821

-> After summing groups of 3: 116101551691211

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[illegible]

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[illegible][illegible]

-> After squaring digits:

[illegible]

Step 2 (current number):

[illegible]

-> After squaring digits:

[illegible]

-> After summing groups of 3:

[illegible]

Step 3 (current number):

[illegible]

-> After squaring digits:

[illegible]

Brutal stable number reached:

[illegible]